

Auckland University of Technology (AUT)

Data Engineering and AI - ENGE707

Progress Report for Phase 1

Data Engineering Foundation for Diabetes Risk Prediction

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Introduction

Type 2 diabetes and prediabetes often stay undiagnosed for a long time, mainly because clinical testing is not always easy to access, and many people do not get tested until symptoms or complications already appear. This project looks at whether a low-cost screening tool, built using self-reported health, lifestyle, and demographic survey data, could help identify at-risk individuals earlier, before clinical testing is even needed.

This kind of screening tool could be useful to a few different groups. Public health agencies could use it to identify communities or population groups that may need more diabetes awareness or screening programs. Healthcare providers could use it as a quick, low-cost first check before recommending clinical testing. Individuals themselves could also benefit, since a simple self-reported tool would let them check their own risk level without needing a doctor's visit first.

The aim of this project is to build an end-to-end data engineering and machine learning pipeline that predicts diabetes/prediabetes risk using CDC Diabetes Health Indicators dataset, and to compare a Learning Classifier System against other machine learning approaches to see which features are most useful for prediction.

This report covers Phase 1 of the project, which focuses on the data engineering foundation: selecting a suitable dataset, acquiring and inspecting it, cleaning and transforming it, and exploring it descriptively. Feature engineering and model building are part of the next phase of this project (Phase 2).

1. Task 1: Problem Definition and Dataset Selection

1.1 Real-World Problem and Motivation

As explained in the introduction, this project investigates whether self-reported health, lifestyle, and demographic data can support early screening for diabetes and prediabetes risk, without relying on clinical testing as the first step. This matters because a large number of type 2 diabetes cases may go undiagnosed until symptoms or complications have already developed, often simply because the person was never tested. A cheap, accessible screening step based on data people already know about themselves (weight, activity levels, general health) could help close that gap by flagging at-risk individuals earlier.

This has practical relevance for a few groups. Public health agencies could use a model like this to identify communities or population groups that may need targeted screening programs. Primary care providers could use it as a quick, low-cost check before recommending clinical testing. Individuals themselves could also use a simplified version to get a sense of their own risk level without needing a doctor's visit first. The BRFSS survey data used here is well suited to this problem because it contains self-reported information about respondents' health, lifestyle, and demographic characteristics (Centers for Disease Control and Prevention [CDC], 2017).

1.2 Dataset Overview

Table 1: Dataset Overview

Attribute	Detail
Name	CDC Diabetes Health Indicators
Source	UCI Machine Learning Repository, Dataset ID 891
Link	https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators
Origin	CDC's 2015 Behavioural Risk Factor Surveillance System (BRFSS)
Size	253,680 records, 21 features + 1 target variable.
Target variable	Diabetes_binary (0 = no diabetes, 1 = prediabetes/diabetes)
Access date	06/08/2026
Licence	CC0: Public Domain, no usage restrictions

1.3 Justifications Against Dataset Selection Criteria

Table 2: Dataset Selection Requirements

Requirement	How this dataset meets it
Tabular, alpha-numeric data (no images)	All 22 columns are numeric survey-response codes; no image data
More than 20 and fewer than 300 features	21 features, within range
Sufficient records for EDA, training, validation, testing	253,680 records, well above minimum
Meaningful classification task	Binary classification: diabetes/prediabetes status
Adequate documentation/metadata	UCI page includes a full variable codebook and BRFSS 2015 provenance
Lecturer approval	Approved after meeting the teaching team (see appendix 1).

2. Task 2: Data Acquisition, Inspection, and Documentation

2.1 Data Acquisition

The dataset was acquired programmatically in Python using the `ucimlrepo` package, which fetches the CDC Diabetes Health Indicators dataset (UCI ID 891) directly from the UCI Machine Learning Repository. The features and target were combined into a single `DataFrame` and saved to `data/raw/diabetes_raw.csv`. Loading the dataset via a reusable script, rather than a manual download, makes the acquisition step reproducible for any team member or marker re-running the pipeline.

2.2 Dataset Structure

The dataset used for this project is the CDC Diabetes Health Indicators dataset, sourced from the CDC's 2015 Behavioural Risk Factor Surveillance System (BRFSS) and hosted on the UCI Machine Learning Repository. It contains a total of 253,680 records and 22 columns, with each row representing one survey participant. Of these, 21 are predictive features and one is the target variable, `Diabetes_binary` (*CDC Diabetes Health Indicators*, 2017). This satisfies the assignment's requirement of more than 20 and fewer than 300 features. The target variable is binary coded, with 0 representing no diabetes and 1 representing diabetes or prediabetes.

2.3 Variable Dictionary

Table 3: Variables Dictionary

Column	Meaning	Type / Range
HighBP	High blood pressure (self-reported)	Binary (0/1)
HighChol	High cholesterol (self-reported)	Binary (0/1)
CholCheck	Cholesterol checks within past 5 years	Binary (0/1)
BMI	Body Mass Index	Numeric (12–98)
Smoker	Smoked ≥ 100 cigarettes in lifetime	Binary (0/1)
Stroke	Ever told had a stroke	Binary (0/1)
HeartDiseaseorAttack	Coronary heart disease or heart attack	Binary (0/1)
PhysActivity	Physical activity in past 30 days (excl. job)	Binary (0/1)
Fruits	Consumes fruit $\geq 1 \times$ /day	Binary (0/1)
Veggies	Consumes vegetables $\geq 1 \times$ /day	Binary (0/1)
HvyAlcoholConsump	Heavy alcohol consumption	Binary (0/1)
AnyHealthcare	Has any health coverage	Binary (0/1)
NoDocbcCost	Couldn't see doctor due to cost (past 12 mo.)	Binary (0/1)
GenHlth	Self-rated general health	Ordinal (1=excellent–5=poor)
MentHlth	Days of poor mental health, past 30 days	Numeric (0–30)
PhysHlth	Days of poor physical health, past 30 days	Numeric (0–30)

DiffWalk	Serious difficulty walking/climbing stairs	Binary (0/1)
Sex	Sex	Binary (0=female, 1=male)
Age	Age bracket (13-level code)	Ordinal (1–13)
Education	Education level bracket	Ordinal (1–6)
Income	Income level bracket	Ordinal (1–8)
Diabetes_binary	Target: diabetes/prediabetes status	Binary (0=no, 1=yes)

2.4 Data Types

All 22 columns were stored using the int64 data type. Most variables are represented as binary or ordinal categories rather than continuous measurements. For example, HighBP uses 0 for no and 1 for yes. Four columns, Age, Education, Income, and GenHlth, also represent ordered categories (brackets) rather than plain integer counts. The meanings and ordering of these coded variables were interpreted using the official BRFSS documentation (CDC, 2016). This is a potential data type issue, since storing ordinal brackets as plain integers risks them being misused in numeric operations that don't make sense (e.g. averaging Age brackets as if they were literal ages). No data types were changed during this task, as the focus at this stage was inspection rather than cleaning; this issue is addressed directly in Task 3.

2.5 Missing Values

No missing values were identified, with all 22 columns containing complete data for the 253,680 records. Although no missing-value treatment is required as a result, it is assumed that this completeness reflects preprocessing already carried out by the dataset's publisher, rather than every participant necessarily answering every original BRFSS survey question.

2.6 Invalid Categories and Inconsistent Coding

The values were inspected in each column to find any invalid categories or inconsistent coding. All binary variables only contained 0 and 1, which matched the dataset documentation. The ordinal variables stayed within their expected ranges:

- GenHlth ranged from 1 to 5.
- Age ranged from 1 to 13.
- Education ranged from 1 to 6.
- Income ranged from 1 to 8.
- MentHlth and PhysHlth ranged from 0 to 30 days.

No invalid category was found. No inconsistent spelling or text-formatting issue appeared because all variables had already been numerically encoded.

2.7 Descriptive Statistics and Value Ranges

BMI ranges from 12 to 98, with a mean of 28.38 and a median of 27, indicating a right-skewed distribution. MentHlth and PhysHlth (poor mental and physical health days, respectively) each range from 0 to 30, with means of 3.18 and 4.24, most respondents report few or no such days, with a smaller number reporting many. The extreme BMI values may be outliers, though this cannot be confirmed from the data alone.

2.8 Duplicate Records

24,206 records (9.54% of the dataset) were found to exactly match an earlier record. The dataset does not include a unique participant identifier, so identical rows cannot be automatically confirmed as accidental duplicates, since most features are broad survey categories with only a small number of possible values, different participants could plausibly give identical answers by chance. Grouping the duplicate rows showed 11,369 distinct duplicate clusters, with a typical cluster size of 2 and a largest cluster of 59 identical rows. These records were documented at this stage but not removed, the decision on how to handle them is addressed in Task 3.

2.9 Assumptions and Limitations

The following assumptions and limitations were identified:

- Each row represents a BRFSS participant.
- Diabetes_binary is the target variable.
- Binary values represent categories, while Age, Education, Income, and GenHlth are ordered categories rather than continuous numeric values.
- Matching rows may represent different participants, since no unique identifier is available.
- Extreme BMI values are outliers rather than confirmed errors.
- The absence of missing values may result from publisher preprocessing.
- The self-reported responses may be affected by recall or response bias.
- The data represents surveyed adults in the United States during 2015 and may not generalise to other countries or populations.

3. Task 3: Data Cleansing and Transformation

Four data quality issues were reviewed during cleaning: duplicate records, BMI outliers, invalid or inconsistent values, and the representation of ordinal variables. No missing values required handling, since Task 2 confirmed the dataset is fully populated.

3.1 Duplicated Records

24,206 duplicated rows (about 9.54% of the dataset) were identified. Closer inspection showed these duplicates form small clusters (median cluster size = 2), scattered across non-adjacent row indices, with the largest clusters made up of common "low-risk" response patterns (e.g. no reported health condition, active lifestyle, normal BMI). Given how many of this dataset's features are binary or low-cardinality, and how large the overall sample is, this pattern looks more like different respondents coincidentally giving identical answers than a data collection or export error.

Therefore, the decision is that the duplicate rows were retained. Removing them would risk disproportionately discarding genuine respondents with common health profiles, which could push the class imbalance even further.

3.2 BMI Outliers

BMI values range from 12 to 98. Both tails were checked: the lowest values (12–13) and highest values (95–98) formed smooth, gradually declining distributions rather than isolated freak values, and were scattered across different rows rather than clustered together. These values are medically extreme but physiologically possible (severe underweight at one end, severe obesity at the other), and there is no sign of data-entry errors, no zero, negative, or absurd values like 300+.

The decision is to retain the BMI outliers rather than remove or cap them, since they likely represent genuine extreme cases in a large population survey, and they are visualised in the EDA stage (boxplot, Task 4) so their effect on the data stays visible rather than hidden.

3.3 Invalid or Inconsistent Values

This was checked as part of Task 2's inspection. Every binary column contained only {0,1}, and every ordinal column (Age, Education, Income, GenHlth) contained exactly the number of category levels defined in the UCI codebook, with no stray or out-of-range codes. No invalid or inconsistent values were found, so no cleaning action was required for this issue.

3.4 Ordinal Variables

Age, Education, Income, and GenHlth are stored as integers but actually represent ordinal bracket/category codes rather than continuous numeric values. Left as plain integers, they risk being misused in numeric operations that don't make sense for them, for example, averaging Age brackets as if they were literal ages. To prevent this, these four columns were converted to pandas' ordered categorical dtype.

3.5 Summary of Cleaning Decisions

Table 4: Summary of Cleaning Decision.

Issue	Finding	Decision	Rationale
Duplicate rows	24,206 rows (9.54%)	Retained	Small, scattered clusters consistent with coincidental overlap in a low-cardinality feature space, not data errors
BMI outliers	Range 12–98	Retained	Smooth distribution tails, no signs of data-entry errors, medically extreme but plausible
Invalid/inconsistent values	None found	No action needed	Confirmed during Task 2 unique-value check across all columns
Ordinal columns (Age, Education, Income, GenHlth)	Stored as int64	Converted to ordered category dtype	Prevents misuse as continuous numeric values
Missing values	None found	No action needed	Dataset confirmed fully populated during Task 2 inspection
Categorical encoding / feature scaling	Not yet needed	Deferred to Phase 2	Phase 1 explicitly excludes feature engineering; encoding/scaling choices depend on the specific models chosen later

No other transformation types, standardising formats, unit conversion, aggregation, or reshaping, were applicable at this stage. The dataset is already consistently numeric with no unit mismatches, and each row already represents one respondent at the correct level of detail. Encoding and scaling were also left out for the same reason as before: Phase 1 excludes feature engineering, so these are deferred to Phase 2.

The cleaned dataset was saved to `data/processed/diabetes_cleaned.csv`.

4. Task 4: Exploratory Data Analysis and Visualisation

4.1 Dataset Validation and Target Distribution

The cleaned dataset from Task 3 was loaded for exploratory data analysis. It contained 253,680 survey responses and 22 variables, including 21 predictors and the Diabetes_binary target. No missing values were found. The 24,206 matching rows and BMI values ranging from 12 to 98 were still present, consistent with the cleaning decisions made and explained in Task 3. Age, education, income, and general health were converted back into ordered categories because CSV files do not retain Pandas category settings.

The target groups were not evenly balanced. Of all respondents, 218,334 (86.07%) were in the no-diabetes group, while 35,346 (13.93%) were in the prediabetes/diabetes group. Because one group was much larger than the other, comparing only the number of cases could be misleading. Therefore, the remaining analysis used percentages to make fairer comparisons between groups of different sizes.

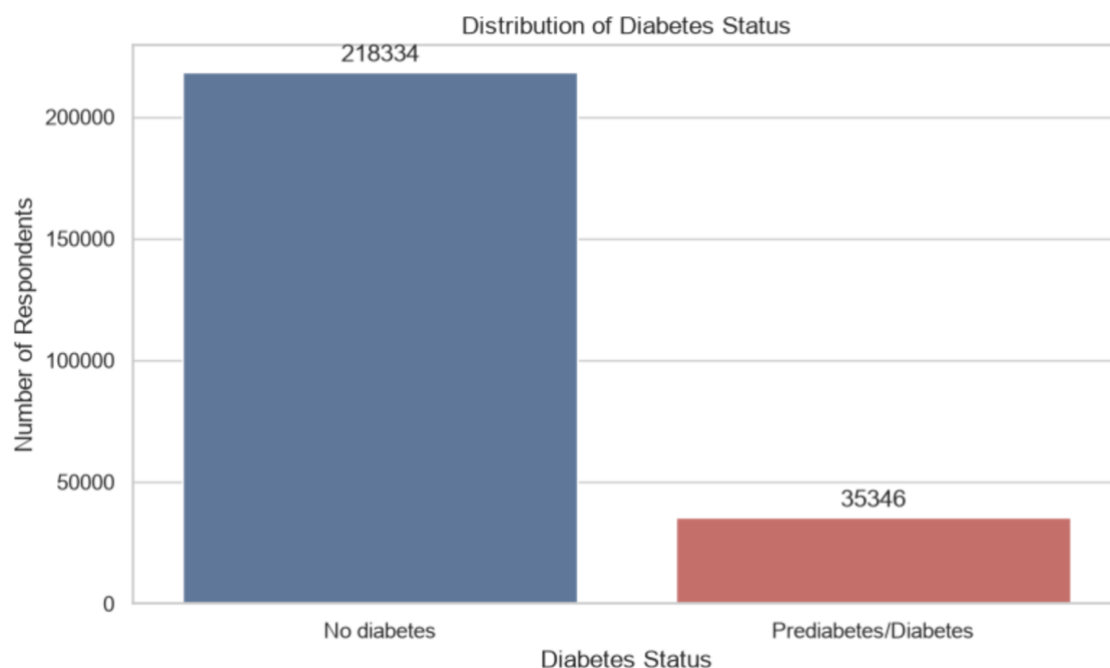


Figure 1: Distribution of Respondents by Diabetes Status.

4.2 BMI distribution and Diabetes Status

BMI had a mean of 28.38 and a median of 27. The mean is higher because the distribution contains a smaller number of very high BMI values. These values stretch the distribution's tail toward the right-hand side indicating a positively skewed distribution.

These outliers were retained because they were physiologically possible and there was no evidence that they resulted from data entry errors.

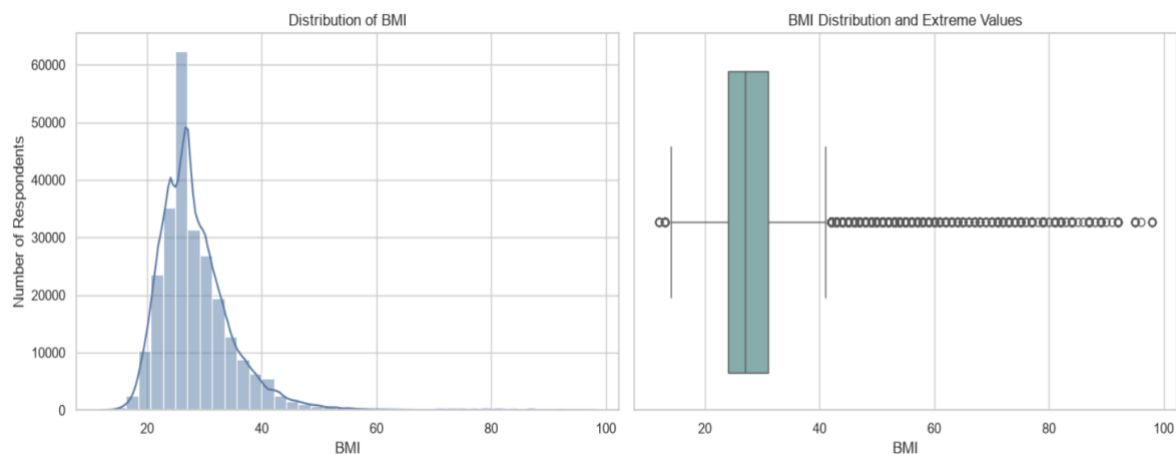


Figure 2: BMI Distribution and Statistical Outliers in the Processed Dataset.

BMI also differed between the target groups. Respondents without diabetes had a mean BMI of 27.81 and a median of 27, whereas respondents classified as having prediabetes or diabetes had a mean BMI of 31.94 and a median of 31. Although the distributions overlapped and BMI could not show and confirm diabetes status by itself, the results displayed an association between higher BMI and diabetes prevalence.

4.3 Health and Demographic Patterns

Diabetes prevalence generally increased across the ordered age brackets. Age was represented using 13 ordered bracket codes, ranging from 18–24 years to 80 years or older (CDC, 2016). It rose from 1.4% in the 18–24 bracket to 21.9% in the 70–74 bracket. The prevalence then declined slightly to 21.3% for ages 75–79 and 18.5% for respondents aged 80 or older. Overall, the pattern indicated substantially higher diabetes prevalence among older respondent groups.

Self-reported general health also showed a clear relationship with diabetes status. Respondents who rated their health as fair or poor had considerably higher diabetes prevalence than those reporting excellent or very good health. Diabetes prevalence was also higher among respondents reporting high blood pressure or high cholesterol. These findings indicate that several existing health conditions and perceptions of poorer health occur more frequently alongside diabetes. However, the comparisons are observational and do not prove that any individual factor directly causes diabetes. In some cases, diabetes could be the reason for certain health conditions or poor health.

4.4 Lifestyle Indicators

Respondents with a smoking history had a diabetes prevalence of 16.3%, compared with 12.1% among those without a smoking history. Respondents reporting physical activity had a rate of 11.6%, compared with 21.1% among those reporting no physical activity. Daily fruit consumers had a rate of 12.9%, compared with 15.8% among non-daily consumers. Daily vegetable consumers had a rate of 13.0%, compared with 18.0% among respondents who did not consume vegetables daily.

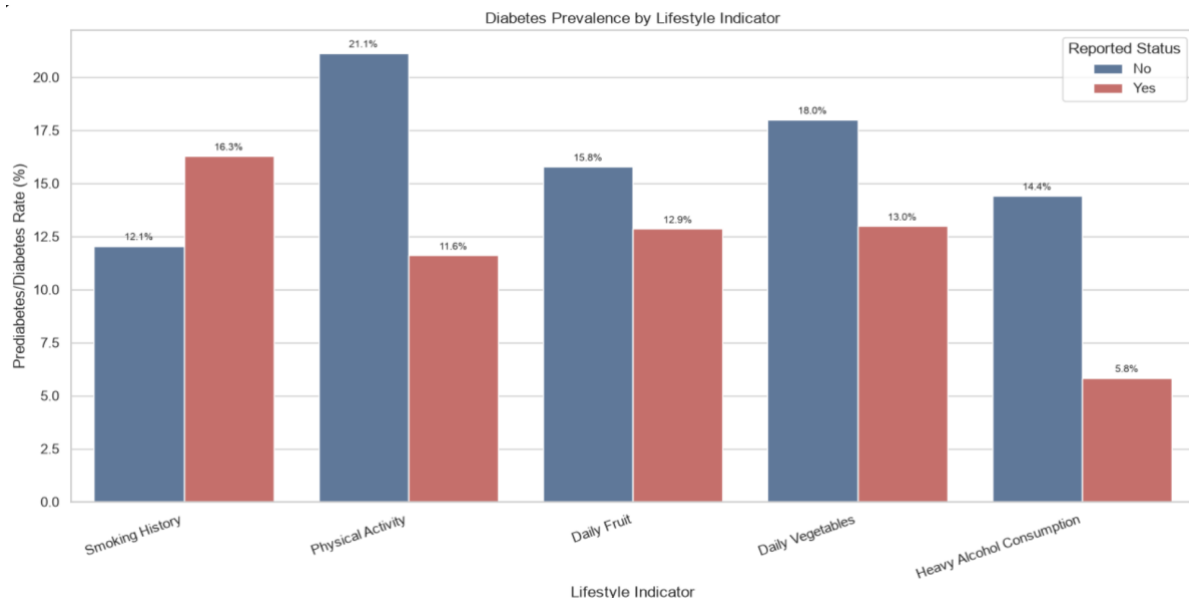


Figure 3: Prediabetes/Diabetes Prevalence by Selected Lifestyle Indicators.

Respondents classified as heavy alcohol consumers show a lower diabetes rate (5.8%) than those who are not (14.4%). This counterintuitive result should not be interpreted as evidence that heavy alcohol consumption is good for preventing diabetes. Different factors such as the respondents' age, health status or the size of the groups may contribute to this pattern.

4.5 Correlation Analysis

Spearman correlation was used to examine selected health, demographic, and lifestyle variables. It was chosen because the analysis included ordered category codes alongside binary and numerical variables.

No single variable had a strong correlation with diabetes status. Poorer general health had the largest positive correlation at 0.29, followed by high blood pressure at 0.26, BMI at 0.23, difficulty in walking at 0.22, and high cholesterol at 0.20. Age and a history of heart disease or heart attack each had a correlation of 0.18.

Income, education, and physical activity had weak negative correlations with diabetes status of -0.16 , -0.12 , and -0.12 respectively.

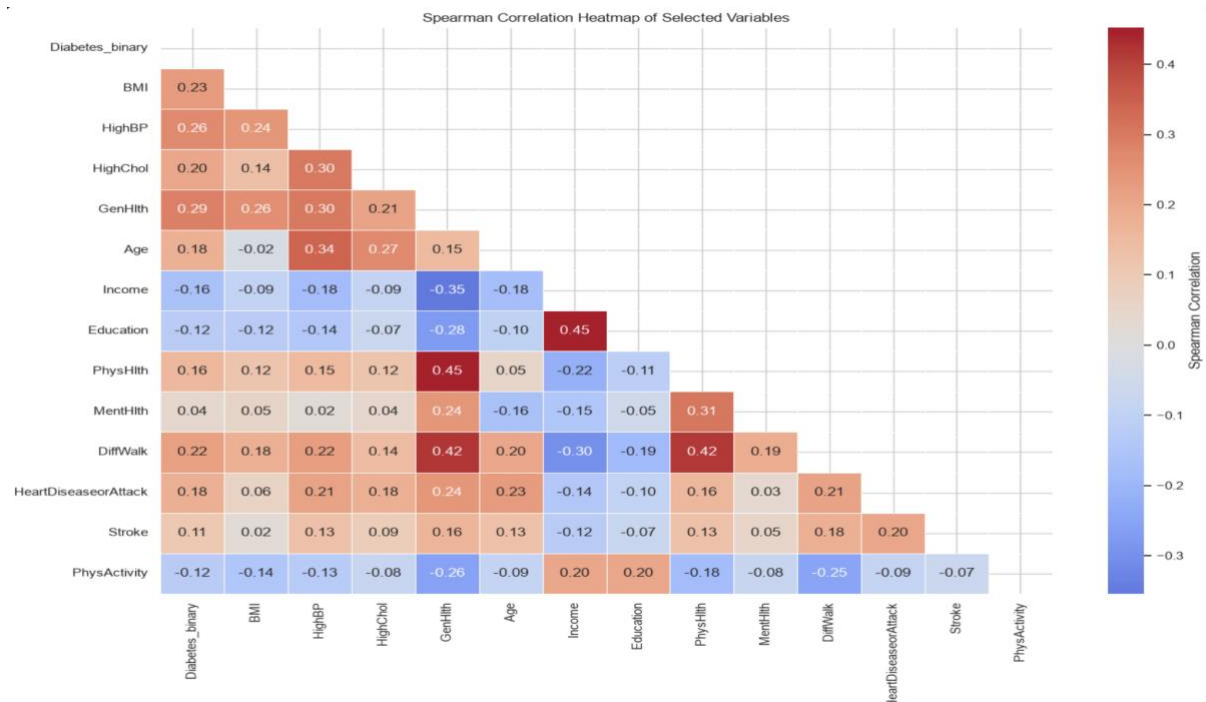


Figure 4: Spearman Correlation Heatmap for Selected Health, Demographic, and Lifestyle Variables.

The strongest relationships between predictors included education and income (0.45) and relationships among general health, poor physical health days, and difficulty in walking (0.42–0.45). These patterns suggest that some socioeconomic and physical health variables contain related information. Correlation shows the direction and strength of relationships but does not establish causation.

4.6 Key Insights and Limitations

This analysis found meaningful relationships between diabetes status and various predictors. However, no single predictor strongly determined prediabetes/diabetes condition, which suggests that diabetes status is related to a combination of health, demographic, and lifestyle characteristics rather than one factor alone.

Several limitations must be considered:

- The dataset relies mainly on self-reported survey responses. Therefore, this may contain recall or reporting bias.
- Age, education, income, and general health are recorded as ordered categories rather than precise continuous measurements.
- Finally, the target classes are imbalanced, which will need to be considered when evaluating future classification models.

All relationships identified during the EDA are observational and should not be interpreted as evidence of direct causal effects.

Conclusion

Phase 1 has established a complete data engineering foundation for this project. The CDC Diabetes Health Indicators dataset was selected, justified against the assignment's criteria, and approved by the teaching team. It was acquired programmatically for reproducibility, thoroughly inspected, and cleaned with clearly justified decisions, retaining duplicate records and BMI outliers as plausible rather than erroneous, and converting ordinal variables to an appropriate categorical type. Exploratory analysis showed a substantially imbalanced target (86.07% no diabetes vs. 13.93% prediabetes/diabetes) and meaningful, though individually weak, associations between diabetes status and factors such as BMI, age, general health, blood pressure, and several lifestyle indicators.

These findings suggest no single variable predicts diabetes status alone, supporting the planned multivariate approach for Phase 2, including the Learning Classifier System comparison. The main considerations carried forward are the class imbalance, the self-reported nature of the data, and the need to encode and scale the ordinal variables during feature engineering.

References

CDC diabetes health indicators [Data set]. (2017). *UCI Machine Learning Repository*.

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Centers for Disease Control and Prevention. (2017, August 11). *2015 BRFSS survey data and documentation*. Retrieved August 6, 2026, from

https://www.cdc.gov/brfss/annual_data/annual_2015.html

Centers for Disease Control and Prevention. (2016, August 24). *Summary matrix of calculated variables in the 2015 data file*. Retrieved August 6, 2026, from

https://www.cdc.gov/brfss/annual_data/2015/summary_matrix_15_version12.html

Appendices

Appendix 1: Lecturer Approval

Subject: ENGE707 Phase 1 dataset approved
Date: Thursday, 6 August 2026 at 1:04:44 PM New Zealand Standard Time
From: Anubha Kalra
To: Khlaf Alshammari

Hi,

Your group's proposed dataset has been approved.

Group name: Irobot

Dataset title: CDC Diabetes Health Indicators

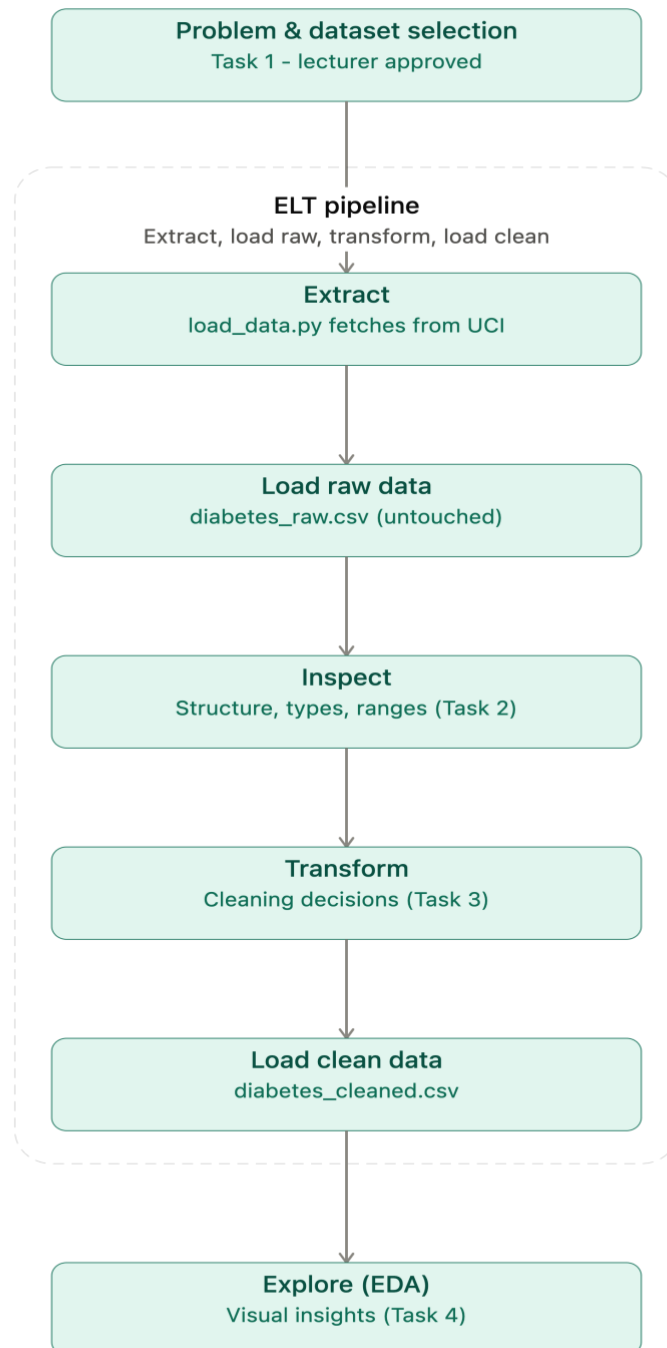
Dataset link:

<https://archive.ics.uci.edu/dataset/891/cdc%2Bdiabetes%2Bhealth%2Bindicators>

You may now proceed with the project.

Regards,
ENGE707 Teaching Team

Appendix 2: Data Pipeline Diagram



Appendix 3: Link to GitHub repository

The following is the link to GitHub repository for this project:

https://github.com/ikhalf0/Data_Engineering_and_Ai